Atmospheric Re-Entry Simulations

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This paper explores spacecraft re-entry dynamics by numerically solving the coupled equations of motion for an exponential atmospheric density model. Thermal loads are quantified using the Sutton-Graves correlation, with a focus on peak heat flux and integral heat load. A parameter study shows how the ballistic coefficient and re-entry angle affect the thermal and structural loads. The work compares a standard ballistic entry with a lifting entry. While a constant lift-to-drag (LOD) ratio results in an undesirable *skip*-trajectory, a variable LOD control law is shown to smooth the descent. Results reveal that this controlled lifting trajectory reduces integral heat loads by approximately 65.5 % and peak deceleration by 15 % compared to the ballistic re-entry.

INTRODUCTION

Earth's atmosphere provides spacecraft the ideal conditions to dissipate their kinetic energy and safely touch down on the surface. However, during re-entry, vehicles experience very high thermal and dynamic loads that require careful mission planning. Simulations are essential for assessing possible flight paths and setting requirements for the structure of re-entry vehicles. This paper explores the characteristics of re-entry trajectories by using a simple set of coupled ordinary differential equations.

The equations of motion are derived by analysis of the forces acting on a spacecraft during re-entry. They consist of a drag force acting opposite to the direction of flight, a lifting force acting perpendicular to the direction of flight and the gravitational force pulling towards the center of Earth. Here, the drag force only changes the magnitude of the velocity while the lift force only changes its direction. Gravity can change both the magnitude and the direction of the velocity vector. Importantly, for a meaningful analysis, the equations have to be transformed into an inertial reference frame, impacting the direction of the velocity vector as the spacecraft changes its position relative to earth. Through a mathematical analysis that is outlined in detail in [1, p. 37], the problem can be described by the following three scalar equations:

$$\dot{h} = -v \cdot \sin(\gamma) \tag{1}$$

$$\dot{v} = -\frac{q}{\beta} + g(h)\sin(\gamma) \tag{2}$$

$$\dot{\gamma} = \frac{1}{v}\left[-\frac{q}{\beta}\left(\frac{L}{D}\right) + \cos(\gamma)\left(g(h) - \frac{v^2}{R_E + h}\right)\right] \tag{3}$$

Here, h is the height above sea level, v the magnitude of the velocity of the spacecraft, $q = \rho v^2/2$ the dynamic pressure, $g(h)$ the height-dependent gravitational acceleration, γ the angle between the velocity vector and the local horizontal and R_E the radius of earth.

SciPy's `solve_ivp` function is used to solve the initial value problem. The integration is terminated once the spacecraft reaches zero height. For all following evaluations an initial height of 100 km and an initial velocity of 7500 m s^{-1} is used, representing a re-entry from low earth orbit.

The thermal analysis is based on the Sutton-Graves-Correlation (Equation 4) that correlates the heat flux q_{heat} onto the surface of the vehicle with its nose radius R, velocity and the local air density. Subsequently, the wall temperature of the spacecraft is estimated by equating the incoming heat flux with the radiated heat flux q_{rad} that is described by the Stefan-Boltzmann law (Equation 5). Since the exponential air density model is based on an isothermal assumption, the ambient temperature is assumed to be at 0 °C so 273.15 K. Furthermore, an emissivity of 0.8 for the heat shield of the spacecraft is assumed.

$$q_{\text{heat}} = 1.74 \times 10^{-4}\,\text{kg}^{0.5}\,\text{m}^{-1}\left(\frac{\rho}{R}\right)^{0.5} u^3 \tag{4}$$

$$q_{\text{rad}} = \varepsilon\sigma\left(T_{\text{wall}}^4 - T_{\text{amb}}^4\right) \tag{5}$$

$\sigma = 5.67\times10^{-8}\,\text{W}\,\text{m}^{-2}\,\text{K}^{-4}$ is the Stefan-Boltzmann constant.

BALLISTIC RE-ENTRY

As a first step, the ballistic re-entry of a capsule with a ballistic coefficient of 400 m s^{-1} and an entry angle of 10° shall be investigated. The ballistic re-entry is characterized by a lift-over-drag (LOD) ratio of zero meaning that the capsule produces no lifting force in the atmosphere. In reality, even simple return capsules are designed to produce at least a small lifting force to distribute the heat-loads during re-entry more evenly.

The simulated trajectory meets all expectations (see Figure 1). In the beginning, the velocity grows by a small amount, as potential energy is converted to kinetic energy. This remains the case until the atmospheric drag exceeds the gravitational acceleration which causes the velocity to reduce steadily. As the capsule enters deeper and deeper into the atmosphere, the drag forces increase substantially through the exponentially growing air density. Eventually, the capsule descends into the deeper

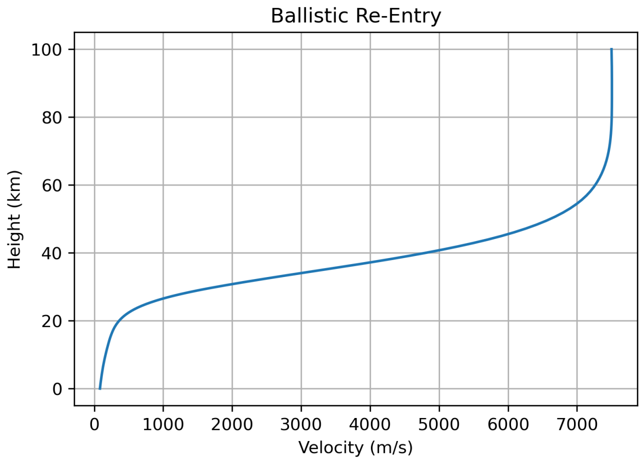

Figure 1. Simulated height over velocity for the ballistic re-entry vehicle.

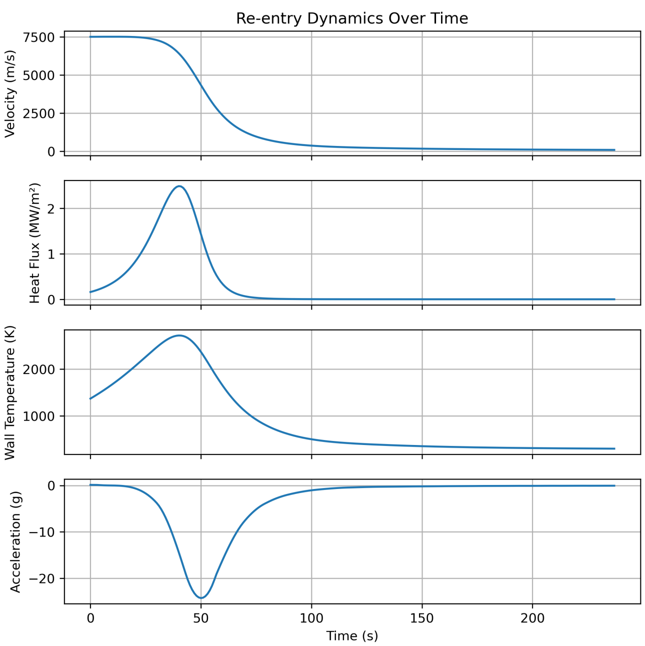

Figure 2. Simulated heat flux, wall temperature, and acceleration over time for the ballistic re-entry vehicle.

parts of the atmosphere at much lower sonic to sub-sonic velocities until it reaches the surface.

Very important for the design of the spacecraft are the thermal and structural loads the vehicle experiences over the duration of the re-entry. Especially the heat and acceleration peaks are important for stress testing of the hardware. Figure 2 shows the evolution of the velocity, heat flux, wall temperature and acceleration over time. One can observe that the peak deceleration of the spacecraft happens about 50 s into the flight. The peak heating, however, occurs slightly before that, at around 40 s into the flight. This makes sense since the $q_{\text{heat}} \propto v^3$, and the velocity decreases over time.

Another interesting parameter is the integral heat load on the spacecraft. For this simulation, a total of $72.95\,\mathrm{MJ\,m^{-2}}$ of thermal energy has to be endured by the vehicle's structure. This is equivalent to 0.65 % of the spacecrafts initial kinetic energy. Thus, only a very small amount of the dissipated energy is absorbed by the vehicle itself. The majority is absorbed by Earths atmosphere.

Finally, a parameter study is conducted to determine what effect the ballistic coefficient and the entry angle have on the integral heat load, maximum heat load, and maximum acceleration. The results are shown in Figure 3. Here, the integral heat load is affected by both parameters, however, by different magnitudes. Smaller entry angles cause a high increase in thermal heat load. An increasing ballistic coefficient seems to have an almost linear impact on the integral heat load. Next, the maximum heat flux is equally increasing towards higher entry angles and higher ballistic coefficients. Lastly, the maximum acceleration only shows a dependency on the entry angle and not on the ballistic coefficient.

LIFTING RE-ENTRY

As a second step, the lift-over-drag ratio is changed to a value of one, i.e. activating the lift force in the set of equations. Now, the trajectory is simulated with the ballistic coefficient $\beta = 50\,\mathrm{kg\,m^{-2}}$ and the entry angle $\gamma = 5°$.

The simulation results are shown in Figure 4. On this plot, one can observe the re-entry vehicle undergoing a *skip*-trajectory (like a stone over water). It takes the spacecraft four attempts until it finally enters the deeper parts of the atmosphere and reaches the surface. This trajectory is suboptimal because it increases the total time spent in the atmosphere and subsequently increases the integral heat load. For a manned spacecraft, this rollercoaster will probably also not make for a pleasant ride.

To remove the *skipping* from the trajectory, a variable LOD value is introduced. The idea is to reduce the LOD when the angle to the horizontal gets shallower. This way, the spacecraft is stopped from moving upwards and the *skip*-trajectory can be avoided. The simplest way to achieve this is by setting

$$\frac{L}{D} = \frac{\gamma}{90°} \qquad \text{for } \gamma \in [0, 90].$$

For negative angles, i.e. velocities that have an upward component, the LOD is set to zero.

Next, a smooth lifting re-entry trajectory is achieved that is very similar in shape to the ballistic re-entry (see Figure 5). At the same time, the other performance parameters show that this lifting re-entry is advantageous compared to the ballistic one. The integral heat

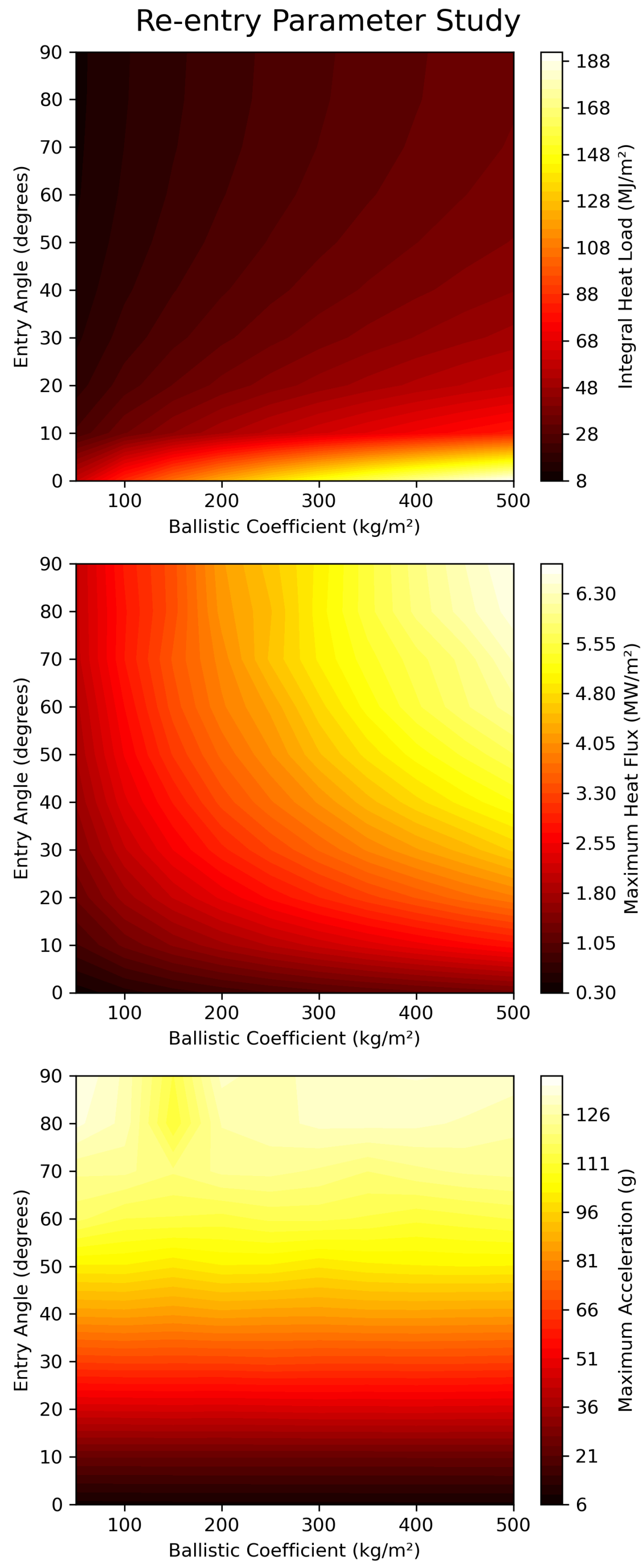

Figure 3. Integral heat load, maximum heat load, and maximum acceleration for different ballistic coefficients and entry angles.

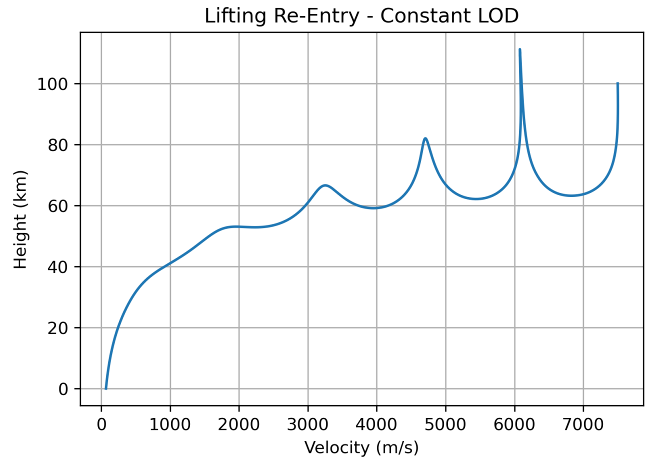

Figure 4. *Skip*-trajectory for the lifting-body of constant $L/D = 1$

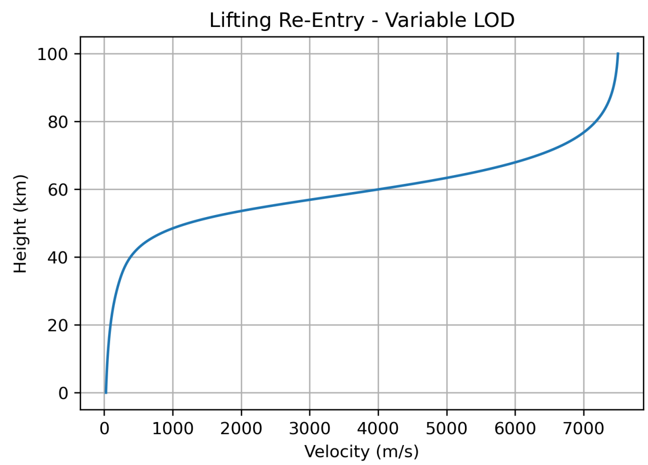

Figure 5. Re-entry trajectory for the spacecraft with a variable LOD

load is 65.5 % smaller at $33.89\,\mathrm{MJ\,m^{-2}}$. Moreover, the maximum heat flux is reduced by a factor of 2.94 at $0.62\,\mathrm{MW\,m^{-2}}$. Lastly, the maximum acceleration is reduced by 15.5 % peaking at 11.6 g.

Finally, the chemical and thermodynamic effects during the duration of the re-entry are described for the example of the re-entry with a variable LOD (see Figure 6). At the start, the spacecraft is approaching from a zone where the atmosphere cannot yet be described as a fluid because of its low density. The comparatively large thermal and chemical relaxation length scales in the upper part of the trajectory result in the presence of significant non-equilibrium effects in the shock layer of the vehicle. At this stage the radiative heating dominates over the convective one, but the total heat flux is still small compared to the rest of the flight. Once the spacecraft approaches a height of 90 km, the characteristic length for the excitation of vibrational states of the atoms becomes comparable to the extent of the vehicle's

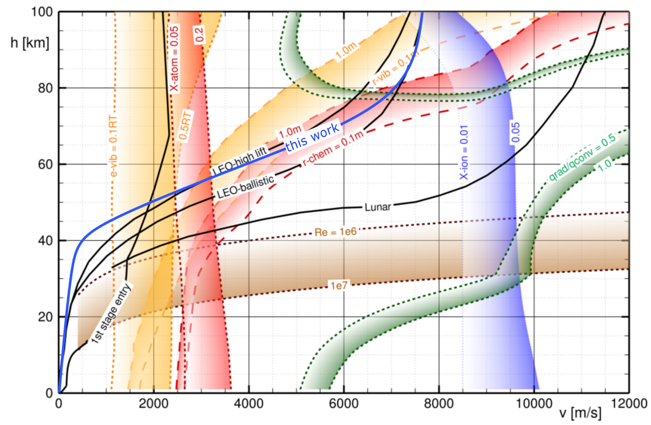

Figure 6. The variable LOD trajectory sketched into a plot taken from [1, p. 46]. Here, the yellow lines represent boundaries of the vibrational excitation, the red lines boundaries of chemical dissociation, and the blue lines ionization boundaries. The green boundaries indicate the boundary between radiative and convective heating, while the brown boundary marks the laminar-turbulent flow transition.

shock layer. These states are now excited right in front of the heat shield. Towards 80 km the characteristic length of the dissociation of molecules also reduces towards relevant scales. A little lower into the atmosphere the density gets high enough for the convective heat flux to dominate over the radiative one. From here until a height of about 60 km, the spacecraft loses about 40 % of its initial velocity. During this shock phase, where both vibrational and chemical excitations are active, the vehicle has to endure the highest thermal and mechanic loads. After this, the characteristic lengths of first the chemical and then the molecular excitation energies grow larger and the shock layer returns into non-equilibrium. Furthermore, since the velocity has already been reduced significantly, the kinetic energy transferred to the particles starts to fall below the excitation thresholds for both mechanisms. Thus, at around 45 km both molecular excitations have reduced to insignificant impact on the overall system. Now, the remaining velocity is directly transferred into the gas's translational and rotational energy (i.e. temperature) and the gas's behave *as usual* again. Between 20 km and the surface, laminar-turbulent flow transitions start to occur.

CONCLUSION

In this paper, different re-entry trajectories were simulated and compared to each other. The results are based on a simplified model of the atmospheric re-entry that is based on several simplifications. For one, an isothermal approach was chosen to describe the atmosphere with an exponential air density distribution. For another, the lift and drag coefficients were either left constant for the whole entry regime, or were subject to large variations implemented through a simplified relation with the angle to the horizontal. To properly model the lift and drag forces, the vehicle's own body and its alignment relative to the direction of flight should be simulated for all flight regimes. Still, the simple model creates realistic re-entry trajectories and provides rough performance characteristics that can be the starting point for more refined simulations.

The comparison of the ballistic and lifting re-entry showed that a lifting re-entry is advantageous when evaluated via the performance characteristics. However, it should be noted that a very ambitious aerodynamic model was chosen that enables the spacecraft to perfectly adjust its LOD during all stages of the flight. In reality, the re-entry surfaces and center of mass would have to be carefully designed to allow for a proper adjustment of the lift and drag coefficients. Anyhow, the simulations accurately explain why almost all modern re-entry vehicles are designed to produce at least a small amount of lift during re-entry.

[1] S. Karl, Fundamentals of space engineering (2025).